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Hierarchical Development of Three Direct-Design Methods for Two-Dimensional Axial-Turbomachinery Cascades

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The direct and inverse blade-design iterations for the selection of isolated airfoils and gas turbine blade cascades are enormously reduced if the initial blade shape has performance characteristics near the desirable ones. This paper presents the hierarchical development of three direct blade-design methods of increasing utility for generating two-dimensional blade shapes. The methods can be used to generate inputs to the direct- or inverse-blade-design sequences for subsonic or supersonic airfoils for compressors and turbines, or isolated airfoils. The examples included for illustration are typical modern turbine cascades, and they have been designed by the direct method exclusively. The first method specifies the airfoil shapes with analytical polynomials. It shows that continuous curvature and continuous slope of curvature are necessary conditions to minimize the possibility of flow separation, and to lead to improved blade designs. The second method specifies the airfoil shapes with parametric fourth-order polynomials, which result in continuous-slope-of-curvature airfoils, with smooth Mach number and pressure distributions. This method is time consuming. The third method specifies the airfoil shapes by using a mixture of analytical polynomials and mapping the airfoil surfaces from a desirable curvature distribution. The third method provides blade surfaces with desirable performance in very few direct-design iterations. In all methods the geometry near the leading edge is specified by a thickness distribution added to a construction line, which eliminates the leading edge overspeed and laminar-separation regions. The blade-design methods presented in this paper can be used to improve the aerodynamic and heat transfer performance of turbomachinery cascades, and they can result in high-performance airfoils in very few iterations.

Introduction

Gas turbine blades are three-dimensional objects operating in a complex flow field. Due to its complexity, the problem is usually reduced to a series of "stacked" two-dimensional problems. Blade design is partly a science and partly an art. The performance of the blades is expressed in the form of some property distribution (such as pressure, velocity, etc.) on the blade surface and in the cascade passage. Three-dimensional turbine blades are generated by "stacking" two-dimensional designs using some rules for: the flow angles from hub to tip at the leading and trailing edge, derived from radial-equilibrium flow conditions; the locus of the centers of gravity of the sections; and the three-dimensional shape of the leading edge and the trailing edge. Compromises in performance must be made to accommodate these three-dimensional constraints.

Demands for improved engine performance require sophisticated methods for blade design, because the performance of the airfoils becomes critical. Blade cascades can be designed by direct or inverse methods. Many definitions for direct and inverse design, analysis and design mode, analysis and optimization mode, are used by other investigators. For the purposes of this paper we will define the terms direct and inverse design as follows.

We define as direct the method in which the designer inputs the geometry of the blade, and the output is the performance from a flow-analysis program. The blade geometry is modified, and the analysis program is used in successive iterations until a desirable performance is obtained. We define as inverse design the various methods in which the designer specifies the performance of the cascade to obtain the geometry, or modifications to a portion of the surface velocity or pressure distribution (driven by expectations of boundary layer behavior), to obtain modifications to the geometry. The latter definition includes what other investigators define as fully inverse, semi-inverse, or simply design methods. Both methods are iterative and are based on the assumption of steady-inflow and steady-outflow conditions (although the flow around the blades during engine operation is inherently unsteady). The direct method is

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laborious; it requires considerable insight and many iterations until an acceptable performance is obtained. On the other hand, the designer has direct control of the various geometric parameters, and structurally or dynamically infeasible blade shapes are excluded before they are analyzed. The inverse method is less laborious, but many (although not all) inverse-design programs must start from a blade geometry generated by a direct method. When inverse-design programs are used, one would like to start the inverse method with a blade shape or performance close to an acceptable design, since this will reduce the number of inverse-design iterations enormously. Therefore the designer needs a good blade shape obtained by a direct method. By the nature of inverse design, its user has less control over the blade shape and on the resulting performance. It may result in blades that are structurally inferior due to stress considerations, or impossible to manufacture. If one decides that some geometric parameters (such as stagger angle) must be changed, one must start from the beginning (direct method) again. The final design is usually obtained by a judicious combination of both methods.

Turbine-Cascade Geometry

In compressor cascades the flow diffuses; the pressure at the outlet is higher than the pressure at the inlet. In subsonic creased, and the area available to the flow should continuously increase, to a maximum area corresponding to the trailing edge of one blade and the pressure surface of the next. Traditionally compressor cascades have been designed by designing individual airfoils, in which thickness distributions are added to either side of a circular, parabolic, cubic, or other-shaped camber line (Horlock, 1958; Wilson, 1984; Moran, 1984; Cumpsty, 1989).

In turbine cascades the flow expands; the pressure at the outlet is lower than the pressure at the inlet. In subsonic cases

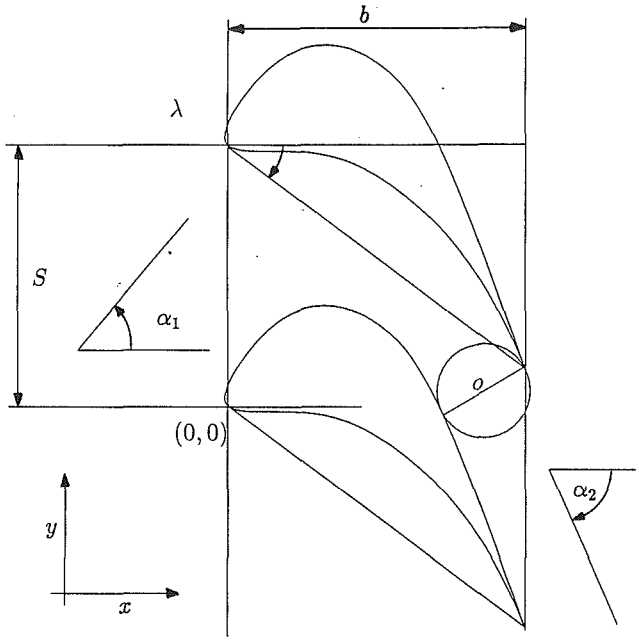


Fig. 1 Geometry of a typical turbine cascade

from cascade inlet to outlet the passage velocity is increased, and the area available to the flow should continuously decrease, to the minimum area corresponding to the circle of diameter o at the throat (see Fig. 1). Although some early turbine cascades were also designed from individual airfoils (Horlock, 1966; Dunham, 1974), traditionally turbine cascades have been designed by manipulating the shape of the passage between successive airfoils, and then closing the shape at the leading edge by specifying a leading edge circle, ellipse, or other ap-

Nomenclature

$\mathbf{A}$ = flow-area distribution along x (Fig. 7b)
a^x, a^y = spline coefficients (Eq. (9))
BC_1, BC_2 = spline derivatives (Eqs. (12), (13), (15), (20))
b = axial chord length (nondimensionally $b = 1$)
c = thickness-distribution coefficients (Eqs. (24))
$\mathbf{C}$ = $1/r$ = curvature (Eq. (7) and Fig. 7b)
$C1, \ldots, C5$ = Bezier-spline points specifying curvature (Fig. 7b)
C_L = tangential-loading coefficient (Eqs. (1), (2))
e = construction-line coefficients (Eq. (23))
fx = spline first derivative (Eqs. (11))
g = body force (Eqs. (4), (5))
K = a constant (spline arclength) (Eqs. (14), (20))
$L(\omega)$ = leading-edge construction lines
M = Mach number
p = pressure (Eqs. (4), (5))
P = points on suction and pressure surfaces
r = local radius of curvature (Eq. (7))
S = tangential spacing of the blades (Figs. 1, 2, 5, 7a)
s = spline parameter (Eq. (9))
sx = spline second derivative (Eqs. (11))
t = time (Eqs. (3), (4), (5))
T = thickness distribution (Eqs. (24))
v_i = velocity component ($i = r, \theta$)
$\mathbf{v}$ = velocity vector (Eq. (6))
(x, y) = Cartesian coordinates
$\mathbf{X}(s), \mathbf{Y}(s)$ = parametric equations for (x, y) (Eq. (9))
$y1, y2$ = surface approximations (Eq. (22))
α = flow angle
β = angle of the blade surface
δ = root-x coefficient for T (Eqs. (24))
δ_1 = boundary layer displacement thickness (Fig. 3c)
λ = stagger angle of the blade
μ = dynamic viscosity (Eq. (6))
ρ = density (Eqs. (3), (4), (5))
τ = stress tensor (Eq. (6))
ϕ = angle of throat diameter with x axis (Fig. 7a)
ω = thickness-distribution parameters (Eqs. (23), (24))

Subscripts

1, 2 = with flow angle and Mach number, inlet and outlet, respectively
1, 2 = with blade point, leading edge, and trailing edge, respectively
a, b, c, d, m = after p or s, point on blade surface
c = construction line
f = point near the leading edge where thickness distribution joins blade surfaces
l = left side boundary condition
p = pressure side
r = right side boundary condition
s = suction side
θ = angle in polar coordinates (r, θ)
κ = construction line, with angle or point (Fig. 9)

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propriate shape (Wilson, 1984; Pritchard, 1985; Korakianitis, 1987a, 1987b, 1989). The flow accelerates around both sides of the cascade, more on the suction and less on the pressure surface. These effects are shown in the performance maps of the sample turbine cascades discussed below.

In recent years the performance of both types of cascade has been improved with the use of inverse-design techniques, otherwise known as prescribed-velocity-distribution (PVD) techniques. Because the pressure is decreased along the turbine stages, any separation that may occur on turbine cascades is likely to reattach, and the fluid will continue its normal flow from turbine inlet to outlet. In compressors the pressure is increased along the stages, and any separation that may occur on compressor cascades may not reattach. This leads to compressor stall and surge. Bad turbine designs will still work, while bad compressor designs will not work. Compressors are harder to design than turbines. However, more attention has been given to compressor designs because of the deleterious effects of bad compressor performance, to the extent that many compressors are more efficient than the "corresponding" turbines (for similar mass-flow rates and pressure ratios). This is because insufficient attention has been paid to turbine-blade design. For the remainder of this paper we will concentrate on turbine cascades, although the methods developed for turbine cascades are directly applicable to compressor cascades and isolated airfoils.

The blade-design sequence depends on the application and the global constraints of the engine. The designer has many choices, but for the purposes of this paper we will refer to the axial-turbine-cascade diagram shown in Fig. 1. The flow is from left to right. One of the early design choices that must be made is the type of velocity diagram. We assume that the designer has chosen the blade row velocity diagram and its parameters such as: the inlet flow angle α_1; the outlet flow angle α_2; and the exit Mach number M_2, in the absolute frame for stators and in the relative frame for rotors. Four additional choices must be made: some specification for the nondimensional tangential spacing S/b between the blades; some specification for the stagger angle λ (the inclination of the blades with respect to the axial direction x); the nondimensional throat diameter o/b; and the trailing-edge thickness.

The stagger angle λ is defined as the angle between the axial direction and the line joining the leading edge of the cascade with the trailing edge point on the concave (pressure) surface. Some guidelines for low-speed turbine blades have been published by Kacker and Okapuu (1982), although the designer has considerable flexibility in choosing λ. The value of λ also dictates whether the cascade is front- or aft-loaded, and different philosophies and guidelines are used by different manufacturers (Korakianitis, 1993a).

The spacing between the blades (which is directly linked to the number of blades in the blade row) is a function of the tangential lift coefficient C_L, defined by:

$$C_L \equiv \frac{\text{tangential aerodynamic force}}{\text{tangential blade area} \times \text{outlet dynamic head}} \quad (1)$$

This expression can be manipulated in a number of ways (for compressible flow, for incompressible flow, accounting for variations in axial-flow velocity, etc.). Here we choose the incompressible-flow derivation (see Wilson, 1984), which reduces C_L to:

$$C_L = 2\frac{S}{b}\cos^2\alpha_2(\tan\alpha_1 - \tan\alpha_2) \quad (2)$$

where b is the axial chord and S is the tangential spacing of the blades. The actual value of C_L depends on the number of blades in the cascade, which dictates S. Typically $0.8 < C_L < 1.2$; earlier designs had lower values of C_L, and in modern cascades C_L has been gradually increasing. It is difficult to design high-efficiency cascades that have both high values of C_L and high values of flow deflection ϵ, where $\epsilon = |\alpha_1| + |\alpha_2|$.

The throat diameter is an extremely important design input, because it dictates the mass flow that can be passed through the cascade, and hence the work that can be delivered by the turbine. A good first approximation is $o/S = \cos\alpha_2$. Experimental figures linking the throat diameter o to the blade spacing S, the throat Mach number M_2, and the curvature of the convex (suction) blade surface near the trailing edge have been published by Ainley and Mathieson (1951), and are used extensively in the early stages of blade designs. The industry has for years relied on empirical relationships based on the Ainley-Mathieson data and on improvements to it (e.g., by Dunham and Came, 1970). These empirical data have been simplified by others for design applications (for example by Wilson, 1984; and Dunham, 1974). This is sufficient for preliminary design. Today's computer programs are more accurate and compute the outlet flow angle for a given geometry. Gostelow (1975) has explained how the result depends not only on the geometry, but also to some extent on the exact location of the point at which the Kutta condition is applied at the trailing edge.

The trailing-edge thickness should be as small as manufacturing considerations would allow to minimize the wake incident on the next blade row, but is also affected by geometric constraints imposed by manufacturing, strength and cooling-slot considerations. In the following figures the trailing edge is a point because we have used inviscid flow solvers to analyze the blade performance. The methods presented enable the designer to specify different trailing-edge points on the suction and pressure surfaces.

In turbine designs the flow of fluid adjacent to the convex (suction) surface is "unguided" on the suction surface downstream of the throat. In successful designs the velocity of the fluid adjacent to the suction surface accelerates from the stagnation point near the leading edge to some point near the throat, and then it decelerates. This region on the suction surface from the throat to the trailing edge is called the region of "unguided diffusion." If the flow decelerates too much then the boundary layer will become thicker and it may separate, the blade wake will be thick, and the resulting cascade will have poor performance (low efficiency). Therefore for high performance we would like the velocity to be continuously and smoothly increasing, from the stagnation point to the trailing edge, along both surfaces. On the suction surface designers like continuously and smoothly increasing velocity from the stagnation point to the point of maximum velocity that occurs near the throat, and minimum and smooth diffusion from the point of maximum velocity to the trailing edge. The Reynolds number of typical production turbine blades is of the order of 5×10^5 based on chord. The flow will probably become turbulent at some point along the surfaces. The smooth condition ensures that the boundary layer is not energized into turbulence unintentionally (although this can be done intentionally by designing a curvature discontinuity at the point where one wants the flow to become turbulent).

Theoretical Dependence on Curvature

Typically the flow in turbomachinery cascades is analyzed by considering Cartesian coordinates (x, y, z), or coordinate transformations in Cartesian systems of coordinates. But the flow needs to *curve* around the blades. The dependence of flow on curvature is seen if one writes the compressible-flow Navier-Stokes equations in the limiting case of cylindrical coordinates (r, θ, z). In particular, since we are interested in "stacking" two-dimensional solutions and geometries, we present here the equations of conservation of mass and momenta in polar coordinates (r, θ), taken from the text by Bird et al. (1960):

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$$\frac{\partial \rho}{\partial t}+\frac{1}{r}\frac{\partial}{\partial r}(\rho r v_r)+\frac{1}{r}\frac{\partial}{\partial \theta}(\rho v_\theta)=0 \tag{3}$$

$$\rho\left(\frac{\partial v_r}{\partial t}+v_r\frac{\partial v_r}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_r}{\partial \theta}-\frac{v_\theta^2}{r}\right)$$

$$=-\frac{\partial p}{\partial r}-\left(\frac{1}{r}\frac{\partial}{\partial r}(r\tau_{rr})+\frac{1}{r}\frac{\partial \tau_{r\theta}}{\partial \theta}-\frac{\tau_{\theta\theta}}{r}\right)+\rho g_r \tag{4}$$

$$\rho\left(\frac{\partial v_\theta}{\partial t}+v_r\frac{\partial v_\theta}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_\theta}{\partial \theta}+\frac{v_r v_\theta}{r}\right)$$

$$=-\frac{1}{r}\frac{\partial p}{\partial \theta}-\left(\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\tau_{r\theta})+\frac{1}{r}\frac{\partial \tau_{\theta\theta}}{\partial \theta}\right)+\rho g_\theta \tag{5}$$

where in the above equations p is the static pressure, and for Newtonian fluids the stress components τ_{rr}, $\tau_{\theta\theta}$, and $\tau_{r\theta}$ are related to the dynamic viscosity μ by the equations:

$$\tau_{rr}=-\mu\left[2\frac{\partial v_r}{\partial r}-\frac{2}{3}(\nabla\cdot\mathbf{v})\right]$$

$$\tau_{\theta\theta}=-\mu\left[2\left(\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}+\frac{v_r}{r}\right)-\frac{2}{3}(\nabla\cdot\mathbf{v})\right]$$

$$\tau_{r\theta}=\tau_{\theta r}=-\mu\left[r\frac{\partial}{\partial r}\left(\frac{v_\theta}{r}\right)+\frac{1}{r}\frac{\partial v_r}{\partial \theta}\right]$$

$$(\nabla\cdot\mathbf{v})=\frac{1}{r}\frac{\partial}{\partial r}(rv_r)+\frac{1}{r}\frac{\partial v_\theta}{\partial \theta} \tag{6}$$

A similar dependence of incompressible flow on curvature is also shown in most fluid dynamics texts in cylindrical coordinates, and Schlichting (1968, p. 112) attributes a set of similar equations for incompressible flow in curvilinear coordinates (x, r, y) to W. Tollmien. The $1/r$ and $1/r^2$ terms in these equations suggest a strong dependence of local velocity on curvature. Once the flow passage has been manipulated until the turbine cascade looks like a turbine cascade, the performance depends on the behavior of the boundary layer, where Eqs. (6) also indicate that the boundary layer is affected by the local radius. Since we require smooth velocity distributions along the blade surfaces (continuous slope of velocity with respect to surface length), the above equations require smooth curvature distributions along the blade surfaces (continuous slope of curvature). Continuous slope of the curvature requires continuous third derivatives, as illustrated in the following two equations for curvature and slope of curvature for $y = f(x)$, $y' = df(x)/dx$, $y'' = d^2f(x)/dx^2$ and $y''' = d^3f(x)/dx^3$.

$$\mathbf{C}=\frac{1}{r}=\frac{y''}{[1+y'^2]^{(3/2)}} \tag{7}$$

$$\mathbf{C}'=\frac{y'''[1+y'^2]-3y'y''^2}{[1+y'^2]^{(5/2)}} \tag{8}$$

The First Blade-Design Method

In the first blade-design method (Korakianitis, 1989) the cascade geometry is specified by five points (excluding the trailing-edge points and the origin) and two slopes on each surface, as shown in Fig. 2. The leading-edge region is generated with two thickness distributions added perpendicularly to two parabolic construction lines. The leading edge geometry is similar in all methods and is described in detail below.

The points on the suction surface are P_{sf}, P_{sa}, P_{sb}, P_{sc}, and P_{sd}. The trailing-edge point P_{s2} (specified by the stagger angle λ and the trailing edge thickness), the angle of the suction surface at the trailing edge β_{s2}, and points P_{sc} and P_{sd} are used to evaluate a third-order analytic polynomial $y_1 = f(x)$ for the blade surface between points P_{sc} and P_{s2}. The value of β_{s2} is 3 to 6 deg smaller (negative values) than the outlet flow angle α_2. Point P_{sf} (user input), β_{sf}, the angle of the suction surface at P_{sf}, and points P_{sa} and P_{sb} are used to evaluate a third-order analytic polynomial $y_2 = f(x)$ for the blade surface between points P_{sf} and P_{sb}. The value of β_{sf} is smaller (positive values) than the inlet flow angle α_1. The segment from P_{sb} to P_{sc} is a fifth-order analytic polynomial $y_3 = f(x)$ evaluated from the six conditions of matching point, first and second derivatives with y_1 and y_2 at P_{sc} and P_{sb}, respectively. This segment could also be a seventh-order polynomial by adding the two additional conditions of matching third derivatives at P_{sb} and P_{sc}, but the seventh-order polynomials frequently result in undesirable inflections on the surfaces. The development of the pressure surface is exactly analogous, using points P_{pf}, P_{pa}, P_{pb}, P_{pc}, P_{pd}, and P_{p2} and angles β_{pf} and β_{p2} (a few degrees larger than the outlet flow angle α_2). The resulting trailing-edge wedge angle is given by the difference of the two slopes of the suction and pressure surfaces at the trailing edge, β_{s2} and β_{p2}, respectively, and it should be between 8 and 14 deg. A disadvantage of this method is that the throat diameter o is an output of the other design inputs.

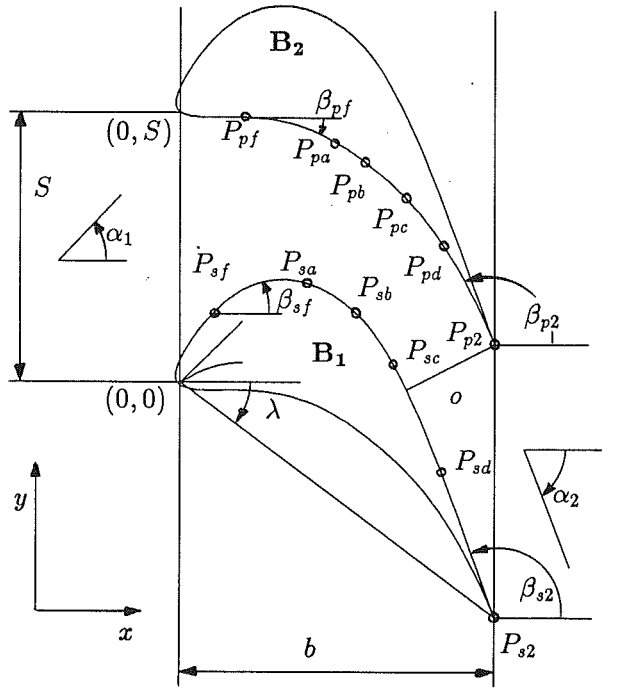
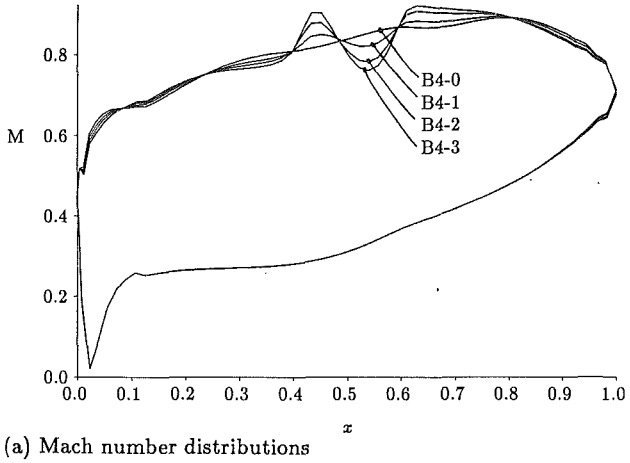


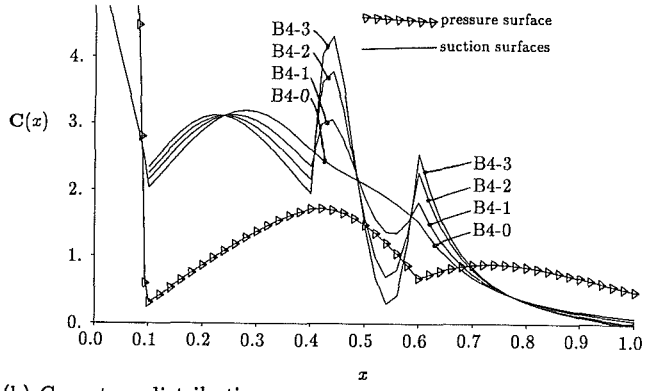
Fig. 2 Parameters used to specify the cascade surfaces in the first blade-design method

Using this method we have designed over 20 turbine cascades of various tangential loadings and deflections. Thirteen of those cascades were used to study unsteady forcing functions in turbomachinery stator-rotor interactions (Korakianitis, 1987a, 1988a, 1988b, 1992a, 1992b). Figure 3 shows the performance of four sample cascades with design inputs very close to each other, as shown in Appendix A (Table 1). The corresponding Mach number contours for cascades B4-0 and B4-3 are shown in Fig. 4. The flows (for all the Mach number distributions in this paper) have been calculated with the steady-flow calculation option of UNSFLO (Giles, 1988). The accuracy of the computations is very high, as shown in tests by Giles (1988) and Korakianitis (1992a, 1993a, 1993b) for numerous steady and unsteady flows in turbomachinery. The boundary layer behavior has been calculated with Rotta's (1971) integral method for compressible two-dimensional flows. In the latter we assumed a Reynolds number based on axial chord of 5.24×10^5, and in all cases transition on both surfaces occurred very near the leading edge. The different inputs result in very small slope-of-curvature discontinuities, which are reflected in both the Mach number distributions and the bound-

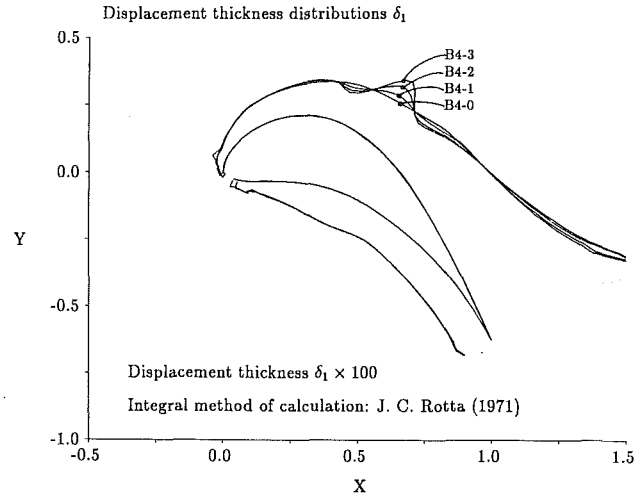
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(a) Mach number distributions



(b) Curvature distributions

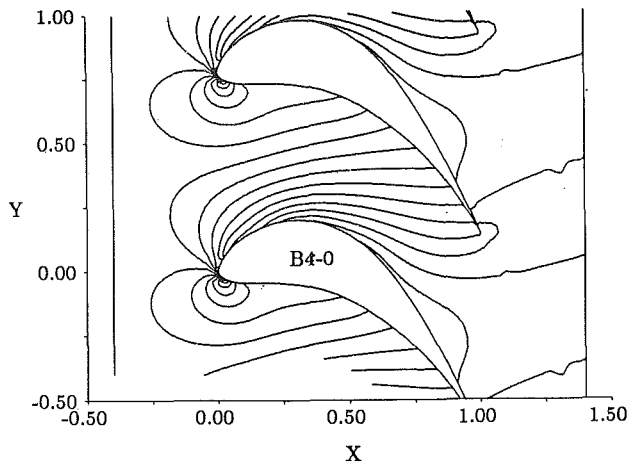


(c) Effect on boundary layers

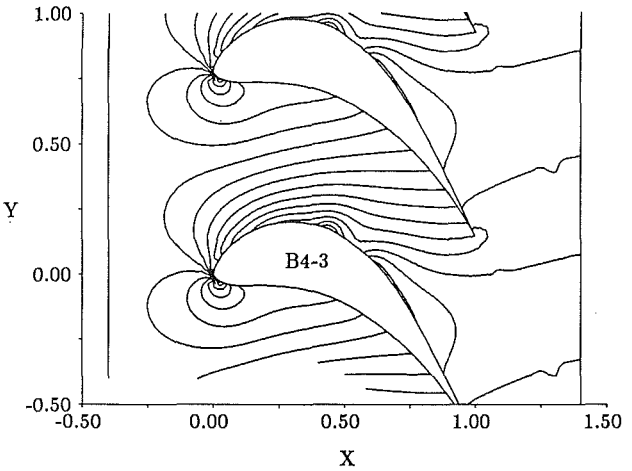
Fig. 3 Surface distributions for cascades B4-0, B4-1, B4-2, B4-3; α_1 = 40 deg, α_2 = −60 deg, C_L = 1.00, λ = −32 deg

ary layer displacement thicknesses. The suction-surface wake from cascade B4-3 (largest slope-of-curvature discontinuity) is thicker than that of cascade B4-0 (no slope-of-curvature discontinuity). The differences in the blade surfaces are so small that they are very hard to see in either Fig. 3(*c*) (where all four blades have been plotted on top of each other), or in Fig. 4.

The first method indicates that even small slope-of-curvature discontinuities may result in unusually loaded cascades (larger curvature spikes have resulted in correspondingly larger Mach number and loading spikes) and thicker wakes. Most techniques for line and surface generation (cubic splines, B splines, Bezier curves, etc.) have continuous first and second derivatives, but not third derivative (slope of curvature) continuity.



(a) For cascade B4-0 (contour increment 0.05)



(b) For cascade B4-3 (contour increment 0.05)

Fig. 4 Mach number contours for cascades B4-0 and B4-3; contour increment 0.05

A slope-of-curvature discontinuity also occurs when a leading-edge circle or ellipse is used to join the leading-edge region. This has resulted in many test and production airfoils (isolated and in turbomachine cascades) that exhibit Mach number and pressure coefficient distribution spikes, dips, and humps of various magnitudes. These occasionally result in unexpected loading distributions along the blade length and in local separation bubbles. Some of these effects are visible in the computational and experimental data published, for example, by: Okapuu (1974), Gostelow (1976), Wagner et al. (1985), Hourmouziadis et al. (1987) (with separation bubble), Baines et al. (1986), Sharma et al. (1992), and possibly Mee et al. (1992a, 1992b) (with separation bubble).

The Second Blade-Design Method

In this method we developed a set of fourth-order parametric splines with third-derivative (slope-of-curvature) continuity at the spline junctions. The nodes of the parametric splines along the suction and pressure surfaces are illustrated in Fig. 5. On the suction surface the nodes are points $\{P_{sf}, P_{sa}, P_{sm}, P_{s2}\}$, and on the pressure surface the nodes are points $\{P_{pf}, P_{pa}, P_{pm}, P_{p2}\}$, although more nodes could be used for each surface. In the following we show the fourth-order parametric splines for n number of nodes $\{P1, P2, \ldots Pn\}$ corresponding to $(n-1)$ spline segments on either surface, where the parameter is s. Let (x_i, y_i) be the Cartesian coordinates of any point P_i on the surface and $(\mathbf{X}_j(s), \mathbf{Y}_j(s))$ be the algebraic description of any spline segment j. The splines take the form:

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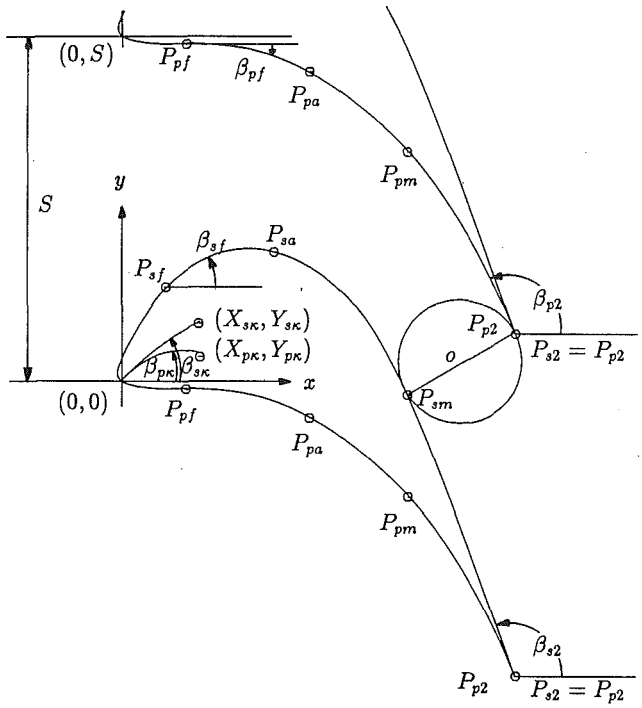


Fig. 5 Parameters used to specify the cascade surfaces in the second blade-design method

$$\begin{aligned}\mathbf{X}_j(s) &= a^x_{j0} + a^x_{j1}s + a^x_{j2}s^2 + a^x_{j3}s^3 + a^x_{j4}s^4 \\ \mathbf{Y}_j(s) &= a^y_{j0} + a^y_{j1}s + a^y_{j2}s^2 + a^y_{j3}s^3 + a^y_{j4}s^4\end{aligned} \tag{9}$$

where $0 \le s \le 1$ in each of the $(n-1)$ segments j, and a are the spline coefficients for $\mathbf{X}(s)$ and $\mathbf{Y}(s)$. In the following we show the derivations applicable to $\mathbf{X}(s)$; the derivation for $\mathbf{Y}(s)$ is exactly analogous. If fx_i and sx_i are the first and second derivatives of $\mathbf{X}(s)$ with respect to s at each of the n nodes i, requiring

$$\begin{aligned}\mathbf{X}_i|_{s=0} &= x_i \\ \mathbf{X}_i|_{s=1} &= x_{i+1} \\ \left.\frac{d\mathbf{X}_i}{ds}\right|_{s=0} &= fx_i \\ \left.\frac{d\mathbf{X}_i}{ds}\right|_{s=1} &= fx_{i+1} \\ \left.\frac{d^2\mathbf{X}_i}{ds^2}\right|_{s=0} &= sx_i\end{aligned} \tag{10}$$

Equation (11) gives the solution for the spline coefficients of the spline segment i in terms of the first and second derivatives at the nodes:

$$\begin{aligned}a^x_{i0} &= x_i \\ a^x_{i1} &= fx_i \\ a^x_{i2} &= sx_i/2 \\ a^x_{i3} &= 4(x_{i+1} - x_i) - fx_{i+1} - 3fx_i - sx_i \\ a^x_{i4} &= 3(x_i - x_{i+1}) + fx_{i+1} + 2fx_i + sx_i/2\end{aligned} \tag{11}$$

The solution for the first and second derivatives at the nodes is found by requiring second- and third-derivative continuity:

$$\left.\frac{d^2\mathbf{X}_i}{ds^2}\right|_{s=1} = \left.\frac{d^2\mathbf{X}_{i+1}}{ds^2}\right|_{s=0}$$

$$\left.\frac{d^3\mathbf{X}_i}{ds^3}\right|_{s=1} = \left.\frac{d^3\mathbf{X}_{i+1}}{ds^3}\right|_{s=0}$$

which for $i = 2, 3, \ldots, n-1$ lead to

$$6fx_{i-1} + sx_{i-1} + 6fx_i - sx_i = 12(x_i - x_{i-1})$$

$$5fx_{i-1} + sx_{i-1} + 6fx_i + sx_i + fx_{i+1} = 4(x_{i+1} + x_i - 2x_{i-1})$$

The last two equations can be manipulated into the matrix form shown in Eq. (12),

$$\begin{bmatrix}
1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & \cdot & \cdot \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & \cdot & \cdot \\
5 & 1 & 6 & -1 & 1 & 0 & 0 & 0 & \cdot & \cdot & \cdot & \cdot & \cdot \\
6 & 1 & 6 & -1 & 0 & 0 & 0 & 0 & \cdot & \cdot & \cdot & \cdot & \cdot \\
0 & 0 & 5 & 1 & 6 & 1 & 1 & 0 & \cdot & \cdot & \cdot & \cdot & \cdot \\
0 & 0 & 6 & 1 & 6 & -1 & 0 & 0 & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot & \cdot \\
\cdot & \cdot & \cdot & \cdot & \cdot & 0 & 0 & 5 & 1 & 6 & 1 & 1 & 0 \\
\cdot & \cdot & \cdot & \cdot & \cdot & 0 & 0 & 6 & 1 & 6 & -1 & 0 & 0 \\
\cdot & \cdot & \cdot & \cdot & \cdot & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
\cdot & \cdot & \cdot & \cdot & \cdot & 0 & 0 & 0 & 0 & 6 & 1 & 6 & -1
\end{bmatrix}$$

$$\begin{bmatrix} fx_1 \\ sx_1 \\ fx_2 \\ sx_2 \\ fx_3 \\ sx_3 \\ \cdot \\ \cdot \\ \cdot \\ fx_{n-1} \\ sx_{n-1} \\ fx_n \\ sx_n \end{bmatrix} = \begin{bmatrix} BC_{x,1}(P_f) \\ BC_{x,2}(P_f) \\ 4(x_3 + x_2 - 2x_1) \\ 12(x_2 - x_1) \\ 4(x_4 + x_3 - 2x_2) \\ 12(x_3 - x_2) \\ \cdot \\ \cdot \\ \cdot \\ 4(x_n + x_{n-1} - 2x_{n-2}) \\ 12(x_{n-1} - x_{n-2}) \\ BC_{x,1}(P_2) \\ 12(x_n - x_{n-1}) \end{bmatrix} \tag{12}$$

from which the values of the first and second derivatives at the nodes can be found. The only remaining unknowns are the right- and left-side boundary conditions

$$\begin{aligned}BC_{lx,1} &= \left.\frac{d\mathbf{X}_1}{dx}\right|_{s=0} \\ BC_{lx,2} &= \left.\frac{d^2\mathbf{X}_1}{ds^2}\right|_{s=0} \\ BC_{rx,1} &= \left.\frac{d\mathbf{X}_{n-1}}{ds}\right|_{s=1}\end{aligned} \tag{13}$$

which are evaluated below.

The first-derivative boundary conditions at the right and left sides of each spline set are related to the angles of the blades at the leading and trailing edge β_{sr}, β_{sl}, β_{pr}, and β_{pl} shown in Fig. 5. These in turn are related to the flow angles α_1 and α_2, and the effect the designer expects that the blade-surface angles will have on the flow angles.

For the first-derivative boundary conditions we can write:

$$\tan\beta = \frac{dy}{dx} = \frac{dy/ds}{dx/ds} = \frac{\sin\beta}{\cos\beta} = \frac{K\sin\beta}{K\cos\beta} \tag{14}$$

so that an obvious choice for the first derivatives of the splines in $\mathbf{X}(s)$ and $\mathbf{Y}(s)$ at the first and last points of the spline sets of the suction or the pressure surfaces is

$$\begin{aligned}BC_{y,1} &= \frac{dy}{ds} = K\sin\beta \\ BC_{x,1} &= \frac{dx}{ds} = K\cos\beta\end{aligned} \tag{15}$$

where K is an arbitrary length chosen to be equal to the arc-length of the appropriate spline segment, and β is the corre-

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sponding blade-surface angle (right for points P_{p2} and P_{s2} and left for points P_{pf} and P_{sf}) from Fig. 5.

Similarly the second derivatives can be expressed as:

$$\frac{d^2y}{ds^2} = K\cos\beta$$

$$\frac{d^2x}{ds^2} = -K\sin\beta \tag{16}$$

Equations (15) and (16) give:

$$\left(\frac{d^2y}{ds^2}\right)\left(\frac{dy}{ds}\right) = -\left(\frac{d^2x}{ds^2}\right)\left(\frac{dx}{ds}\right) \tag{17}$$

The curvature and second derivatives are also related by:

$$\frac{d^2y}{dx^2} = \mathbf{C}\left[1+\left(\frac{dy/ds}{dx/ds}\right)^2\right]^{3/2} = \frac{(d^2y/ds^2)(dx/ds) - (d^2x/ds^2)(dy/ds)}{(dx/ds)^3} \tag{18}$$

Equations (17) and (18) are solved for the second derivatives with respect to s.

$$\frac{d^2x}{ds^2} = -\mathbf{C}\left(\frac{dy}{ds}\right)\left[\left(\frac{dx}{ds}\right)^2+\left(\frac{dy}{ds}\right)^2\right]^{1/2}$$

$$\frac{d^2y}{ds^2} = \mathbf{C}\left(\frac{dx}{ds}\right)\left[\left(\frac{dx}{ds}\right)^2+\left(\frac{dy}{ds}\right)^2\right]^{1/2} \tag{19}$$

which finally give the second-derivative boundary conditions at points $P1$, where $P1$ in Eqs. (20) denotes the "left" points P_{pf} and P_{sf}:

$$BC_{x,2}(P1) = \left.\frac{d^2x}{ds^2}\right|_{P1} = -K^2\left|\mathbf{C}\right|_{P1}\sin\beta|_{P1}$$

$$BC_{y,2}(P1) = \left.\frac{d^2y}{ds^2}\right|_{P1} = -K^2\left|\mathbf{C}\right|_{P1}\cos\beta|_{P1} \tag{20}$$

The above procedure is applied for the splines in x and y for the suction and the pressure surfaces. Point P_{sm} is special, and it is instrumental for the suction-surface design. It is constrained to move on a circle with center the trailing edge of the top blade and radius o. Out of all the points that lie on the circle we choose the ones that correspond to splines with slopes perpendicular to line $P_{p2}P_{sm}$ at point P_{sm}.

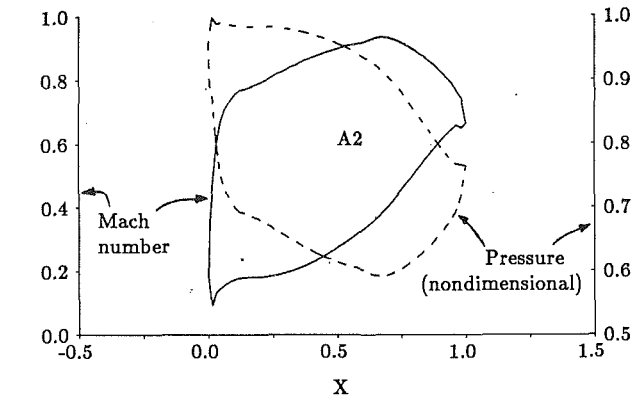
Using this method, we constructed a program that varies the locations of the nodes while searching for curvature distributions with desirable characteristics and continuously decreasing area distributions (Korakianitis and Pantazopoulos, 1993c). The performance of a sample cascade with inlet flow angle of 0 deg, outlet flow angle of −60 deg, and tangential-loading coefficient $C_L = 1.00$ is shown in Fig. 6, and the corresponding blade-design parameters are included in Appendix B (Table 2).

These fourth-order parametric splines may have additional applications, such as smooth automobile surface generation for aesthetically pleasing reflections. Designing new turbine blades with the second blade-design method was a CPU-intensive proposition. After designing six different cascades we developed the third-design method.

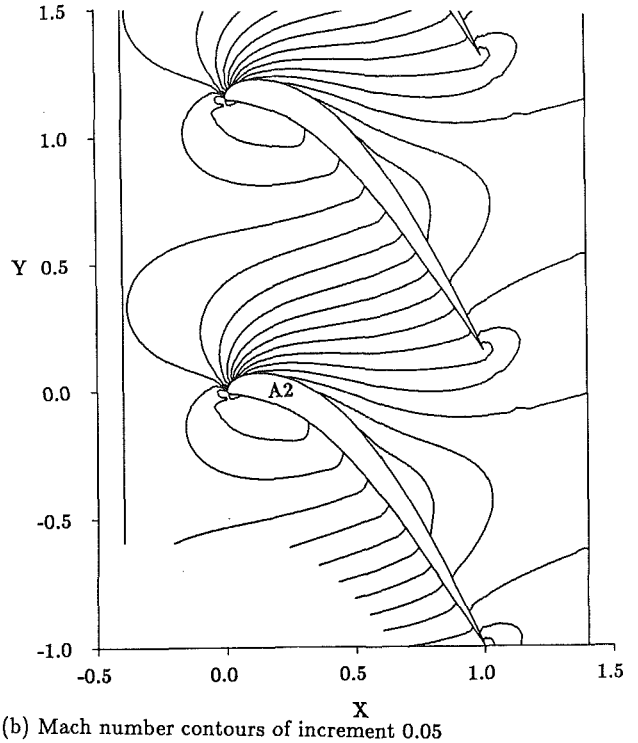
The Third Blade-Design Method

In the third design method part of the blade surface is an analytic polynomial, and part of the blade surface is evaluated by mapping a desirable curvature distribution on the (x, y) plane, as shown in Figs. 7(*a*) and 7(*b*).

The trailing edge region of the suction surface is specified by a third-order polynomial $y = f(x)$, evaluated using four conditions: the location of P_{s2}; the blade angle β_{s2} at the trailing



(a) Surface Mach number distributions



(b) Mach number contours of increment 0.05

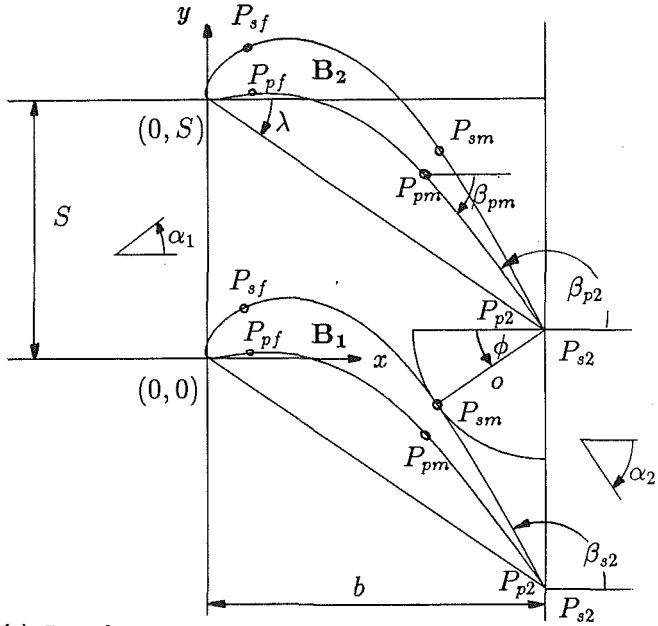
Fig. 6 Performance of a cascade design with the second blade-design method; surface Mach number distribution and Mach number contours of increment 0.05

edge; and the angle ϕ of the throat-circle diameter with the x axis. The latter specifies two conditions: the location of point P_{sm}, and the tangent of the third-order polynomial at that point, equal to $(\pi/2+\phi)$. The analytic polynomial has continuous first and second derivatives. With these we compute the curvature of the resulting blade surface $\mathbf{C}_s(x)$ (using Eq. (7)) between points P_{sm} and P_{s2}, and plot it between points $C4_s$ and $C5_s$ in Fig. 7(*b*).

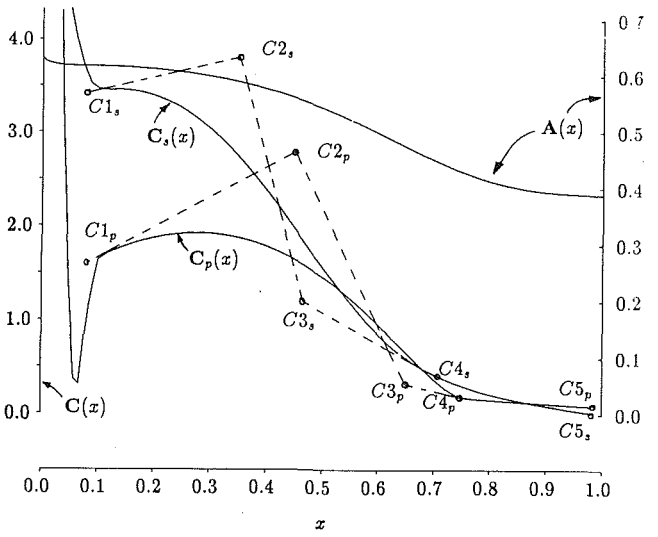
The trailing edge region of the pressure surface is specified by a third-order polynomial $y = f(x)$, evaluated using four conditions: the location of P_{p2}; the blade angle β_{p2} at the trailing edge; the (x, y) coordinates of a point P_{pm} on the surface; and the angle β_{pm} of the blade surface at point P_{pm} with the x axis. With the first and second derivatives of this polynomial we compute the curvature of the resulting blade surface $\mathbf{C}_p(x)$ (using Eq. (7)) between points P_{pm} and P_{p2}, and plot it between points $C4_p$ and $C5_p$ in Fig. 7(*b*).

The slopes of the curvatures $\mathbf{C}_s(x)$ and $\mathbf{C}_p(x)$ at points $C4_s$ and $C4_p$, respectively, are computed and become inputs for the rest of the design. A gradually increasing curvature distribution in the trailing edge regions results in mechanically and aerodynamically feasible blade shapes.

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(a) Cascade geometry



(b) Corresponding curvature distribution

Fig. 7 Parameters used to specify the cascade surfaces in the third blade-design method

The design is converted into an effort of specifying a curvature distribution for the shape of the blade surfaces $\mathbf{C}_s(x)$ and $\mathbf{C}_p(x)$. In order to ensure slope-of-curvature continuity, the curvature from points P_f to points P_m is specified using a Bezier spline in curvatures using the following conditions (see Fig. 7(*b*)). On the curvature of the suction surface we specify points $C1_s$, $C2_s$, $C3_s$, and $C4_s$. Point $C4_s$ is already specified from the corresponding value of curvature evaluated from the trailing-edge third-order polynomial. The x location of point $C3_s$ is variable, but the value of curvature there is evaluated such that line $C3_sC4_s$ is tangent to the line of curvature at point $C4_s$. Point $C2_s$ is user specified. Point $C1_s$ is specified at an x location corresponding to P_f, or at an x location just ahead of P_f. Since the slope of the Bezier curve is tangent to the line of knots at its ends, the tangency condition at point $C4_s$ ensures slope-of-curvature continuity from $C1_s$ to $C5_s$ (from P_{sf} to P_{s2}). Similarly points $C1_p$, $C2_p$, and $C3_p$ are specified on $\mathbf{C}_p(x)$. $\mathbf{A}(x)$ in Fig. 7(*b*) is the area distribution between blades $\mathbf{B}_1$ and $\mathbf{B}_2$, measured from the pressure surface of blade $\mathbf{B}_2$ to the suction surface of blade $\mathbf{B}_1$.

Using central differences, Eq. (7) is written as:

$$\mathbf{C}_i = \frac{CF1/CF2}{CF3}$$

$$CF1 = 2\left(\frac{y_{i+1}-y_i}{x_{i+1}-x_i} - \frac{y_i-y_{i-1}}{x_i-x_{i-1}}\right)$$

$$CF2 = (x_{i+1}-x_{i-1})$$

$$CF3 = \left\{\left[\frac{1}{2}\left(\frac{y_{i+1}-y_i}{x_{i+1}-x_i} + \frac{y_i-y_{i-1}}{x_i-x_{i-1}}\right)\right]^2 + 1\right\}^{3/2} \quad (21)$$

From the above equation given (x_{i-1}, y_{i-1}), (x_i, y_i), x_{i+1} and $\mathbf{C}_i$ we can compute y_{i+1}. This can be done by manipulating Eq. (21) into a sixth-order linear algebraic equation in y_{i+1}, or by a numerical solution. We have implemented a regular-falsi solution for y_{i+1} using Eq. (21). This is solved starting from points P_m and progressing explicitly to points P_f. The Bezier spline is iteratively manipulated until the slope and the y location of the blade surfaces at points P_f, and the shape of the curvature distributions are acceptable.

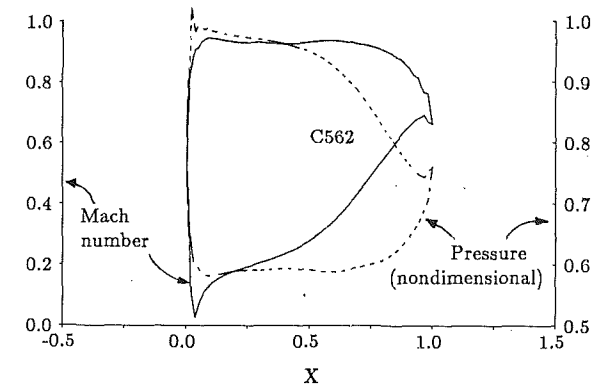
In some cases trying to make the blade surfaces pass with desirable slopes through desirable P_f points become tedious for the inexperienced in the method. For those cases we provide an alternative solution. The designer specifies the (x, y) location of point P_f. A parabola passing through points P_f, P_m and the next point on the blade surface in the trailing edge region is used as a first approximation $y1(x)$ to the blade surface. Equation (21) is manipulated into:

$$[y2_{i+1} - 2y2_i + y2_{i-1}] = \Delta x^2 \mathbf{C}_i \left[\left(\frac{y1_{i+1}-y1_{i-1}}{2\Delta x}\right)^2 + 1\right]^{3/2} \quad (22)$$

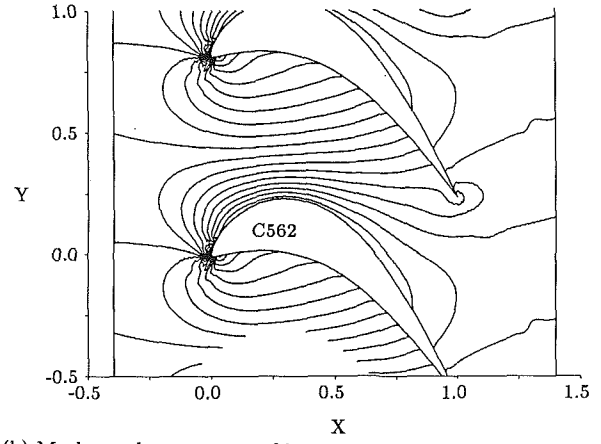
Equation (22) is written as a tridiagonal matrix of the form $[A]\ \{y2\} = \{B(y1)\}$. Points P_m and the input points for P_f are incorporated as boundary conditions in the first and last rows of the right-hand side. The system of equations is solved for $y2_i$ with an LU decomposition. The values of $y1_i$ on the right-hand side are set equal to the previous solution for $y2_i$, and the process is repeated until the values of $y1_i$ and $y2_i$ converge to the same solution (after about twelve iterations). The converged blade surface in the final $y2_i$ ($y2(x)$) passes through points P_f and P_m, but in the general case results in a spike in curvature at point P_m. The curvature-input points $C1$, $C2$, and $C3$ are iteratively manipulated until the spike is minimized, and then having found the curvature-inputs that will make the blade surface pass through point P_f, the designer returns to specifying the blade surface by solving Eq. (21).

Using this method we can produce the geometry of a continuous-slope-of-curvature turbine cascade in a few seconds, and compute its expected flow performance on a relatively coarse grid (40 by 10 grid points in one passage) in about ten DECstation-3100 CPU minutes. Successive direct-design iterations result in changes in the blade shape, curvature, and loading distributions, which we can direct towards front-loaded, midloaded, or aft-loaded performance. The final blade-shape and performance changes are computed on finer grids (100 by 40 grid points in one passage). These design iterations result in about one high-performance cascade per day (Korakianitis, 1993a). High performance means smoother Mach number and pressure distributions than typical ones published in current literature (such as the examples in (Sharma et al., 1992)). These turbine geometries have been used to study the effects of stagger angle on the type of loading distribution (front- versus aft-loaded), and time-dependent stator-rotor interactions in turbomachinery flow fields (Korakianitis, 1992c, 1993a, 1993d). The performance of a high-deflection highly loaded sample cascade with inlet flow angle of 50 deg, outlet flow angle of -60 deg, and tangential-loading coefficient $C_L = 1.20$ is shown in Fig. 8, and the corresponding blade-design parameters are included in Appendix C (Table 3).

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(a) Surface Mach number distributions



(b) Mach number contours of increment 0.05

Fig. 8 Performance of a cascade designed with the third blade-design method

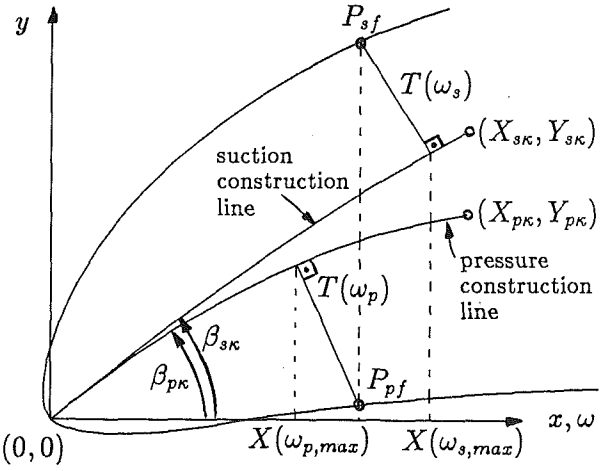


Fig. 9 Parameters used to specify the leading edge geometry

Leading Edge Geometry

The leading edge geometry is designed by specifying thickness distributions added perpendicularly to parabolic construction lines that pass through the leading edge, as shown in Fig. 9. The parabolic construction lines are specified by the angle of the construction line at the origin β_κ and the (x, y) location of the origin (0, 0), and of one additional point (X_κ, Y_κ) on the corresponding (suction and pressure) construction line. Other similarly smooth forms of construction lines could also be used. The perpendiculars from points P_f on the parabolic construction lines $L(\omega)$ define a parameter ω along x as follows (see Fig. 9):

$$\omega = x$$

$$L(\omega) = e_0 + e_1\omega + e_2\omega^2 \tag{23}$$

where the constants e_0, e_1, and e_2 are evaluated from the conditions of the parabolic construction lines. For the suction line we add a thickness distribution above the suction-side parabolic construction line, and for the pressure line we add a thickness distribution below the pressure-side parabolic construction line. The thickness distributions are of the form:

$$T_s(\omega) = \delta_s\sqrt{\omega} + c_{s1}\omega + c_{s2}\omega^2 + c_{s3}\omega^3 + c_{s4}\omega^4$$

$$T_p(\omega) = \delta_p\sqrt{\omega} + c_{p1}\omega + c_{p2}\omega^2 + c_{p3}\omega^3 + c_{p4}\omega^4 \tag{24}$$

where δ are specified by the designer. The coefficients c_1, c_2, c_3, and c_4 are derived with the conditions that the absolute values, first, second, and third derivatives of the thickness distribution and of the blade surface to the right of point P_f *with respect to the parabolic construction line* are equal. This ensures curvature and slope-of-curvature continuity at points P_f. The value of the parameter δ can be used to specify thicker or thinner leading edges, which would affect the thickness of the blade in the vicinity of the leading edge, and therefore its design-point and off-design point operating performance. The origin (0, 0) itself is a singular point. In order to achieve first-derivative continuity at that point, the angle of the parabolic construction lines should be equal for the suction and pressure surfaces ($\beta_{s\kappa} = \beta_{p\kappa}$, and approximately equal to the inlet flow angle α_1).

This type of thickness distribution has been used in the second and third blade-design methods. In the first blade-design method the thickness distributions had point, first, and second derivative continuity with the blade surface, but not third derivative continuity ($c_{s4} = c_{p4} = 0$). The effect of the resulting curvature discontinuity is seen at $x = 0.10$ in Fig. 3(*a*). These small discontinuities are not evident in the performance of the cascades designed with the second and third blade-design methods.

Conclusions

The hierarchical development of three blade-design methods has been presented. Small slope-of-curvature discontinuities result in Mach-number or pressure-coefficient distribution spikes or humps. These may introduce regions of local acceleration and deceleration, separation, or undesirable loading distributions along the length of the blades. Fourth-order parametric splines have been introduced and used to design high-performance cascades, but their use proved CPU intensive. The third blade design method uses a mixture of analytic polynomials and a mapping of a desired curvature distribution on the (x, y) plane to specify the geometry of the cascades. The third method has been used with success to design highly loaded high-performance cascades. These methods, and particularly the third, can be used to design subsonic or supersonic blades for compressors or turbines, or isolated airfoils, but the discussion in this paper is limited to high-Mach-number subsonic-exit turbine blades. The methods permit the user to specify the leading edge by two thickness distributions around two independent construction lines, in a manner that avoids the overspeed regions near the leading edge (because it does not employ the usual blending in of the curvatures near the "leading-edge circle or ellipse"). It is found that the blade performance, judged by the shape of the surface Mach number distribution, is extremely sensitive to small changes of the surface geometry and of the curvature distribution of the airfoils. These methods ensure that the boundary layer, which is usually turbulent along the blade surfaces, is not excited unintentionally. The smooth curvature and Mach number distributions also enable the user to design lower-loss higher-loaded cascades for modern applications in airborne transportion. Continuous-slope-of-curvature surfaces should be

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used wherever the aerodynamic, heat transfer, hydrodynamic, or aesthetic performance, appropriately defined in each case, is critical.

Acknowledgments

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APPENDIX A

Table 1 Parameters specifying blades B4-0, B4-1, B4-2, B4-3

Blade Parameter	B4-0	B4-1	B4-2	B4-3
α_1	40	40	40	40
α_2	-60	-60	-60	-60
C_L	1.00	1.00	1.00	1.00
λ	-32	-32	-32	-32
$\beta_{s\kappa}$	35.00	35.00	35.00	35.00
$XS_{s\kappa}$	0.1000	0.1000	0.1000	0.1000
$YS_{s\kappa}$	0.0650	0.0660	0.0660	0.0660
δ_s	0.1300	0.1300	0.1300	0.1300
β_{sf}	33.50	33.90	34.20	34.50
$x(P_{sf})$	0.1000	0.1000	0.1000	0.1000
$y(P_{sf})$	0.1476	0.1476	0.1476	0.1476
$x(P_{sa})$	0.2000	0.2000	0.2000	0.2000
$y(P_{sa})$	0.1965	0.1965	0.1965	0.1965
$x(P_{sb})$	0.4000	0.4000	0.4000	0.4000
$y(P_{sb})$	0.1956	0.1956	0.1956	0.1956
$x(P_{sc})$	0.6000	0.6000	0.6000	0.6000
$y(P_{sc})$	0.0644	0.0644	0.0644	0.0644
$x(P_{sd})$	0.8000	0.8000	0.8000	0.8000
$y(P_{sd})$	-0.2200	-0.2200	-0.2200	-0.2200
β_{s2}	-65.70	-65.33	-64.80	-64.50
$\beta_{p\kappa}$	35.00	35.00	35.00	35.00
$XS_{p\kappa}$	0.1000	0.1000	0.1000	0.1000
$YS_{p\kappa}$	0.0500	0.0500	0.0500	0.0500
δ_p	0.3000	0.3000	0.3000	0.3000
β_{pf}	2.00	2.00	2.00	2.00
$x(P_{pf})$	0.1000	0.1000	0.1000	0.1000
$y(P_{pf})$	-0.0350	-0.0350	-0.0350	-0.0350
$x(P_{pa})$	0.2000	0.2000	0.2000	0.2000
$y(P_{pa})$	-0.0340	-0.0340	-0.0340	-0.0340
$x(P_{pb})$	0.4000	0.4000	0.4000	0.4000
$y(P_{pb})$	-0.0635	-0.0635	-0.0635	-0.0635
$x(P_{pc})$	0.6000	0.6000	0.6000	0.6000
$y(P_{pc})$	-0.1660	-0.1660	-0.1660	-0.1660
$x(P_{pd})$	0.8000	0.8000	0.8000	0.8000
$y(P_{pd})$	-0.3400	-0.3400	-0.3400	-0.3400
β_{p2}	-61.00	-61.00	-61.00	-61.00

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APPENDIX B

Table 2 Parameters specifying blade A2

Blade Parameter	A2
α_1	0.00
α_2	-60.00
C_L	1.00
o/S	0.5000
λ	-45.00
$\beta_{s\kappa}$	0.00
$XS_{s\kappa}$	0.2000
$YS_{s\kappa}$	0.1500
δ_s	0.1000
$\mathbf{C}(P_{sf})$	3.5319
β_{sf}	12.7050
β_{s2}	-65.0000
$x(P_{sf})$	0.1000
$y(P_{sf})$	0.0699
$x(P_{sa})$	0.2856
$y(P_{sa})$	0.0255
$x(P_{sm})$	0.5772
$y(P_{sm})$	-0.2403
$x(P_{s2})$	1.0000
$y(P_{s2})$	-1.0000
$\beta_{p\kappa}$	0.0000
$XS_{p\kappa}$	0.2000
$YS_{p\kappa}$	-0.0500
δ_p	0.1000
$\mathbf{C}(P_{pf})$	1.4700
β_{pf}	-16.4891
β_{p2}	-56.5033
$x(P_{pf})$	0.1000
$y(P_{pf})$	-0.0279
$x(P_{pa})$	0.4500
$y(P_{pa})$	-0.2471
$x(P_{pm})$	0.7000
$y(P_{pm})$	-0.5546
$x(P_{p2})$	1.0000
$y(P_{p2})$	-1.0000

APPENDIX C

Table 3 Parameters specifying blade C562

Blade Parameter	C562
α_1	50.00
α_2	-60.00
C_L	1.00
o/S	0.5000
λ	-30.00
$\beta_{s\kappa}$	50.0000
$XS_{s\kappa}$	0.2000
$YS_{s\kappa}$	0.2000
δ_s	0.1000
β_{s2}	-64.5000
ϕ	37.0000
$x(C1_s)$	0.0800
$y(C1_s)$	2.9000
$x(C2_s)$	0.3000
$y(C2_s)$	4.5000
$x(C3_s)$	0.6500
$\beta_{p\kappa}$	50.0000
$XS_{p\kappa}$	0.2000
$YS_{p\kappa}$	0.1400
δ_p	0.1000
β_{p2}	-53.0000
β_{pm}	-47.0000
$x(P_{pm})$	0.7500
$y(P_{pm})$	-0.2725
$x(C1_p)$	0.0800
$y(C1_p)$	1.3800
$x(C2_p)$	0.4000
$y(C2_p)$	2.5500
$x(C3_p)$	0.6000

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